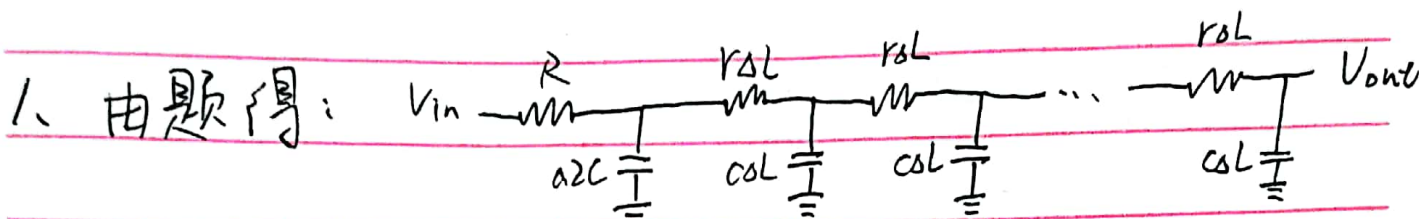


数字VLSI (作业5)

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$$T_D = a_2 R C + \frac{r_0 C L^2}{2}, \text{ 其中, } r = \frac{R}{l_0}, 3C < C < 11C$$

$$\therefore t_{pmax} = a_6 \cdot a_2 R C + a_{38} \frac{R}{l_0} \cdot 11 C L^2 = a_{138} R C + a_{418} R C L^2$$

$$t_{pmin} = a_6 \cdot a_2 R C + a_{38} \frac{R}{l_0} \cdot 3 C L^2 = a_{138} R C + a_{114} R C L^2$$

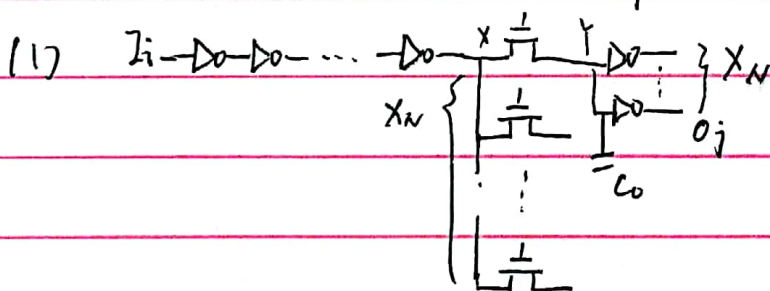
2. $R = a_1 \Omega/\mu m \cdot 500 \mu m = 50 \Omega < 5 \Omega_0 = 250 \Omega$

$$2.5 t_f = 2.5 L/u = 2.5 \times 500 \times 10^{-6} / 1.5 \times 10^8 = 8.33 ps > 2.3 ps$$

$\therefore$ 应采用有损传输线模型

3. ~~传输管的交叉开关: $t_p \propto O(N^2)$~~

~~(2) 二选一的MUX的交叉开关: $t_p \propto O[(\log_2 N)^2]$~~



总延时 $t = t_{i \rightarrow v} + t_{v \rightarrow y}$, 其中 $t_{i \rightarrow v}$ 为反相器延时

$$t_{v \rightarrow y} = a_6 R_d (C_{in} + N C_g) + (R_d + R_t) (N C_0 + N C_g) \cdot a_6$$

$$= a_6 R_d (C_{in} + 2 N C_g) + a_6 R_t (N C_0 + N C_g)$$

$$\text{则 } t = t_1 + t_2 = t_{i \rightarrow v} + a_6 R_d (C_{in} + 2 N C_g) + a_6 R_t (N C_0 + N C_g)$$

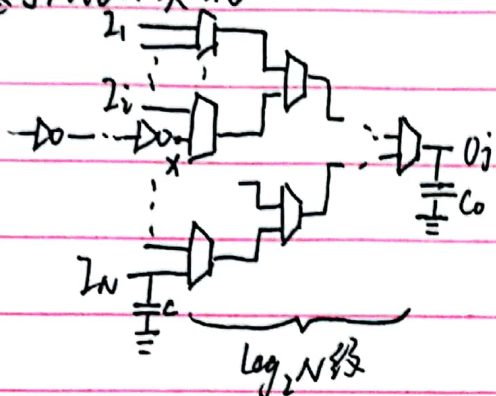
优化 t_1 : $F = \frac{2 N C_g}{C_{g1}}$, 其中 C_{g1} 为第1级反相器输入电容, 优化后

$$\bar{t}_1 = (k+4k)t_{p0}, \quad k = \log_4 F = \log_4 (2N C_0 / C_{g1}) \Rightarrow \bar{t}_1 = 5t_{p0}k$$

$$\bar{t}_1 = 5t_{p0} [\log_4 (2N) + \log_4 (C_0 / C_{g1})]$$

$$\therefore \bar{t}_1 = O(\log_4 N), \quad t_2 = O(N), \quad \therefore t = O(N)$$

(2) 基于MUX实现



设优化后整个路径共 k 级,

有 $F = \frac{C_0}{C_{g1}}$, MUX 的本征延时为 P_{max}

逻辑努力 g_{max} , $\because$ 存在分支, 分

支努力 $b = N$, 则路径努力

$$H = GBF = g_{max}^{\log_2 N} \cdot N \cdot \frac{C_0}{C_{g1}}$$

$$k = \log_4 H = \log_4 (g_{max}^{\log_2 N} \cdot N \cdot \frac{C_0}{C_{g1}})$$

$$= \log_2 N \cdot \log_4 g_{max} + \log_4 N + \log_4 F$$

路径归一化延时 $D = (k - \log_2 N) + \log_2 N P_{max} + 4k$

$$= 5k - \log_2 N + \log_2 N \cdot P_{max}$$

$$= 5(\log_2 N \log_4 g_{max} + \log_4 N + \log_4 F) + (P_{max} - 1)\log_2 N$$

$$= O(\log_2 N)$$

$$\therefore t = O(\log_2 N)$$